

LESSON 3-3 • SESSION 1 • PRACTICE • FORM A

Equivalent Ratios and Tables

Learning target – I can build a ratio table by scaling both rows by the same factor. · 6.RP.A.3a

REMEMBER

- **Equivalent ratios** make the same comparison. Multiply or divide **both** terms by the **same** number.
- In a **ratio table**, each column is one pair. Find the factor, then use it on the other row.
- Adding the same number to both terms does **not** keep the ratio.

WORKED EXAMPLE

A mix uses 5 scoops of powder for every 2 cups of water. How much water for 15 scoops?

Step 1 Find the factor: $15 \div 5 = 3$.

Step 2 Multiply the water by 3 too: $2 \times 3 = 6$.

Step 3 Check: $15 : 6 \div 3 = 5 : 2$.

Answer 15 scoops go with 6 cups of water

WORDS equivalent ratios (*razones equivalentes*) · ratio table (*tabla de razones*) · scale factor (*factor de escala*)

PRACTICE

show your work · label units

1. Find the missing number.

a. $4 \times \underline{\quad} = 20$

b. $\underline{\quad} \times 3 = 19.2$

c. $2.3 \times \underline{\quad} = 18.4$

d. $\underline{\quad} \div 3 = 5$

e. $24 \div \underline{\quad} = 6$

f. $23.1 \div \underline{\quad} = 4.2$

2. **Lemonade Recipe.** A recipe uses 3 lemons for every 2 cups of syrup. Complete the table, then find the syrup for 18 lemons.

Hint Count by 3s on the top row and by 2s on the bottom row.

Lemons	3	6	9	12
Syrup (cups)	2			

Answer _____ cups for 18

3. **Pancake Batter.** Pancakes need 4 cups of mix for every 3 cups of milk. Complete the ratio table.

Hint Find what 4 was multiplied by, then use it on the 3.

Mix (cups)	4	8		20
Milk (cups)	3		9	

4. **Class Supplies.** A class gets 18 notebooks for every 6 students. Scale down to complete the table.

Hint Divide both rows by the same number each time.

Notebooks	18	9		3
Students	6		2	

5. **Sugar to Flour.** A recipe uses 15 cups of flour for every 7 cups of sugar. How much sugar goes with 45 cups of flour?

Hint $45 \div 15$ gives the scale factor.

Show your work.

Answer _____ cups

6. **Pencils to Pens.** A store sells pencils to pens in a ratio of 4.5 to 9. How many pens go with 27 pencils?

Show your work.

Answer _____ pens

7. **Bike Ride.** A cyclist rides 13.5 miles every 1.5 hours. How far does she ride in 4.5 hours?

Hint $1.5 \text{ hours} \times 3 = 4.5 \text{ hours}$.

Show your work.

Answer _____ miles

8. **Ribbons to Bows.** A craft kit uses 2.5 yards of ribbon for every 6 bows.

- a. Write two equivalent ratios that use only whole numbers. _____
- b. Is $4.5 : 8$ equivalent to $2.5 : 6$? Explain. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Deshawn’s Work.** Deshawn says $4 : 6$ and $6 : 8$ are equivalent because he added 2 to each number. Explain his mistake. Then write a ratio that really is equivalent to $4 : 6$.

TEST PRACTICE

choose the best answer

10. Which ratio is equivalent to $2 : 5$?

- A $4 : 7$
- B $8 : 20$
- C $5 : 2$
- D $6 : 16$

11. To keep a ratio the same in a ratio table, you –

- A change only the top row
- B add the same number to both rows
- C multiply both rows by the same number
- D change only the bottom row

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

57.6 Q	12 C	6.4 F	40.5 R	17 X	21 T	54 O	37 Z	5.5 A
#1b	#1f	#2	#5	#6	#7			

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____